

Current System Limitations

| S.No | Limitation                                             | Impact                                             | Priority to Address |
|------|--------------------------------------------------------|----------------------------------------------------|---------------------|
| 1    | Single Flask dev-server process (no load balancing)    | Limits concurrent request throughput               | High                |
| 2    | No authentication on the analyst dashboard             | Anyone with the URL can submit predictions         | Medium              |
| 3    | Static model trained once, not retrained automatically | Model may drift as fraud patterns evolve over time | Medium              |

Scalability Plan

| S.No | Scalability Aspect | Current State                       | Proposed Upgrade / Solution                                               |
|------|--------------------|-------------------------------------|---------------------------------------------------------------------------|
| 1    | User Load          | Single local Flask process          | Deploy behind Gunicorn/Nginx or via IBM Watson ML™'s managed scaling      |
| 2    | Data Storage       | No persistent DB; stateless scoring | Add a lightweight database for prediction/audit logs as usage grows       |
| 3    | Performance        | ~45 req/sec on local dev server     | Horizontal scaling via IBM Watson ML deployment or containerized replicas |
| 4    | Security           | No auth layer                       | Add login-based access control and HTTPS enforcement                      |

Future Roadmap

| Phase   | Planned Feature / Enhancement                                      | Target Timeline | Expected Impact                                             |
|---------|--------------------------------------------------------------------|-----------------|-------------------------------------------------------------|
| Phase 2 | Add authentication & role-based access for analysts                | +1 month        | Restrict access to authorized bank staff                    |
| Phase 3 | Automated model retraining pipeline on new transaction data        | +2 months       | Keeps the model current as fraud patterns shift             |
| Phase 4 | Batch/CSV upload support and analyst review dashboard with logging | +3 months       | Supports real bank-scale daily transaction review workflows |